

Zscaler Digital Experience: Fundamentals



Welcome to Zscaler Digital Experience (ZDX) Fundamentals! Explore the full-featured solution that meets the needs of buyers in the Digital Experience Monitoring market. Dive into ZDX's background and critical capabilities. Gain an understanding of how ZDX solves customers' pains, delivers quantifiable benefits, and supports key initiatives that are important to buyers through its unique differentiation.

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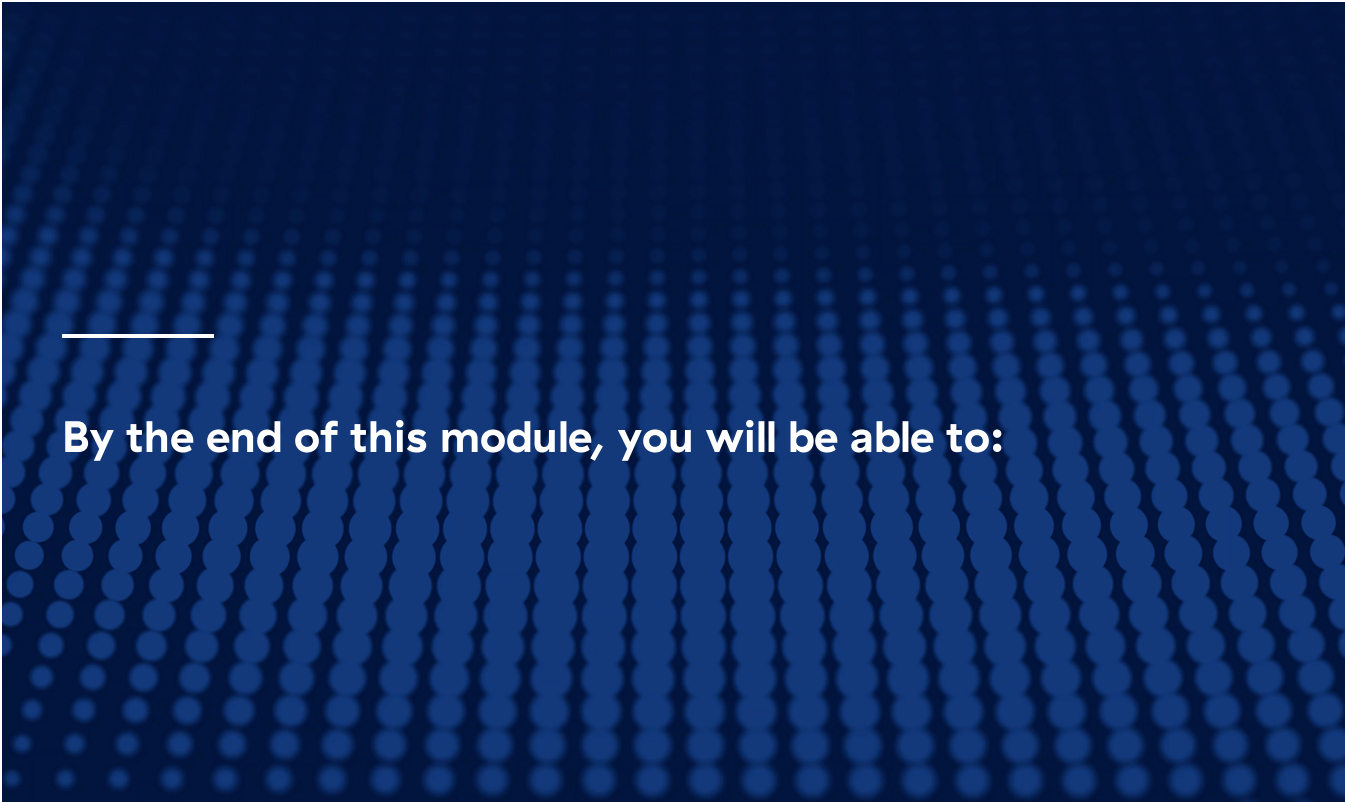
Introduction



In this module, you will explore one of the core product offerings Zscaler has in place to meet the needs of buyers in the industry: **Zscaler Digital Experience (ZDX)**. You will dive into the product's critical capabilities, benefits, pains it solves, and the initiatives that are driving adoption. You will also gain insight into the unique differentiating value the product brings.

CONTINUE

Learning Objectives



By the end of this module, you will be able to:

1

Explain what Zscaler Digital Experience (ZDX) is, why it exists, and what it does

2

Describe the value of ZDX, including the challenges it solves for & benefits it provides

3

Identify the key initiatives and unique points of differentiation

LET'S GET STARTED

What Is ZDX?



“Zscaler Digital Experience (ZDX) is a cloud-native service, part of the world’s largest security cloud that analyzes, troubleshoots, and resolves user experience issues.”

Before diving into the critical capabilities, value, and differentiation that the offering provides, it's first important to understand what ZDX is and why it was built as part of Zscaler's Zero Trust Exchange. Watch the video below to get started.



ZDX Background

Click here for a transcript of the video. —

Hi everyone, I'm Travis Rekowski, Sales Specialist for Zscaler's Digital Experience, and today we're going to be learning about the fundamentals of Zscaler Digital Experience or ZDX, one of Zscaler's core Zero Trust Exchange platform offerings.

To begin we will discuss the background of ZDX and how it came to be. Zscaler Digital Experience helps IT teams monitor Digital Experiences from the end user perspective to optimize performance and rapidly fix offending application, network, and device issues.

ZDX is a cloud-native service that was built on the world's largest security cloud. The Zscaler Zero Trust Exchange, and leverages the scale of the globally distributed Zscaler Cloud to integrate with our existing product lines, including Zscaler Internet Access and Zscaler Private Access. This core Zscaler offering makes it easy to continuously monitor SaaS and private application performance from every single user from every device they're coming from.

But how did we get here? Why was ZDX created? Digital transformation creates monitoring gaps. The rapid adoption of cloud and mobile initiatives within organizations and a shift to work from anywhere has introduced new monitoring challenges for IT teams.

Applications such as Office 365 and Salesforce are moving out of the data center and into the cloud. They're being accessed by a hybrid remote workforce meaning IT teams no longer control

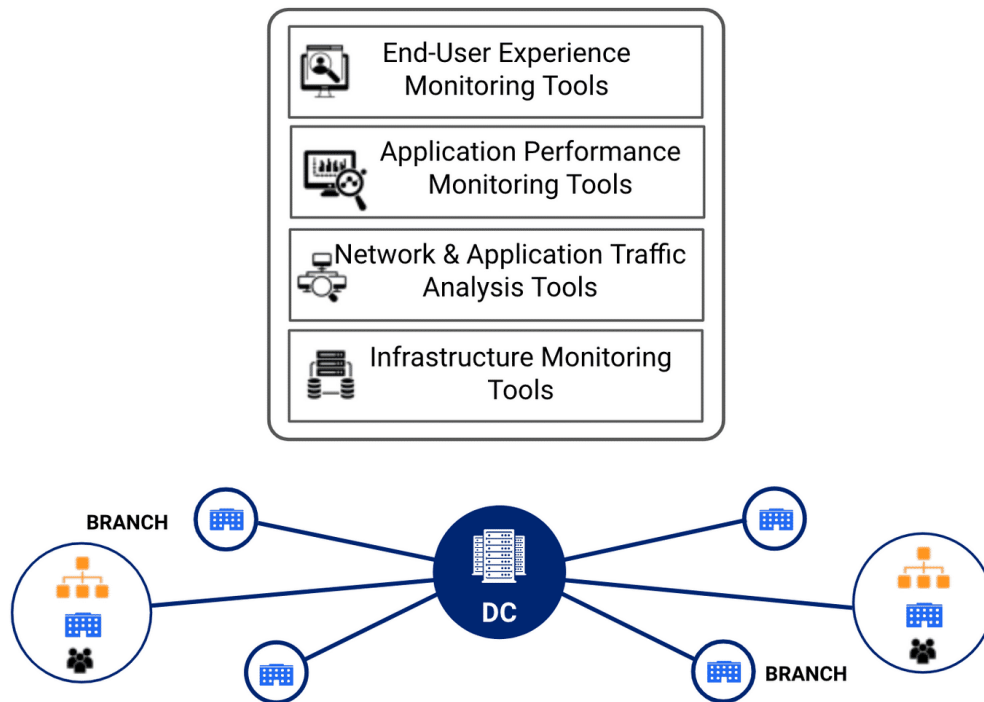
the underlying infrastructure and technology stack and lose end-to-end visibility into the user experience and user performance issues arising from SaaS or cloud application outages, home wifi issues, network path outages, or network congestion, are not easily isolated and diagnosed.

CONTINUE

Monitoring Needs Have Changed With Cloud and Work From Anywhere

In the video, you heard that traditionally when organizations owned the security and networking stacks, applications and infrastructure lived in their own data centers and within their perimeter.

BEFORE



IT Teams were able to instrument monitoring tools wherever needed that provided them with the desired visibility, whether it was the data path to the application from users, or within the applications and data center itself.

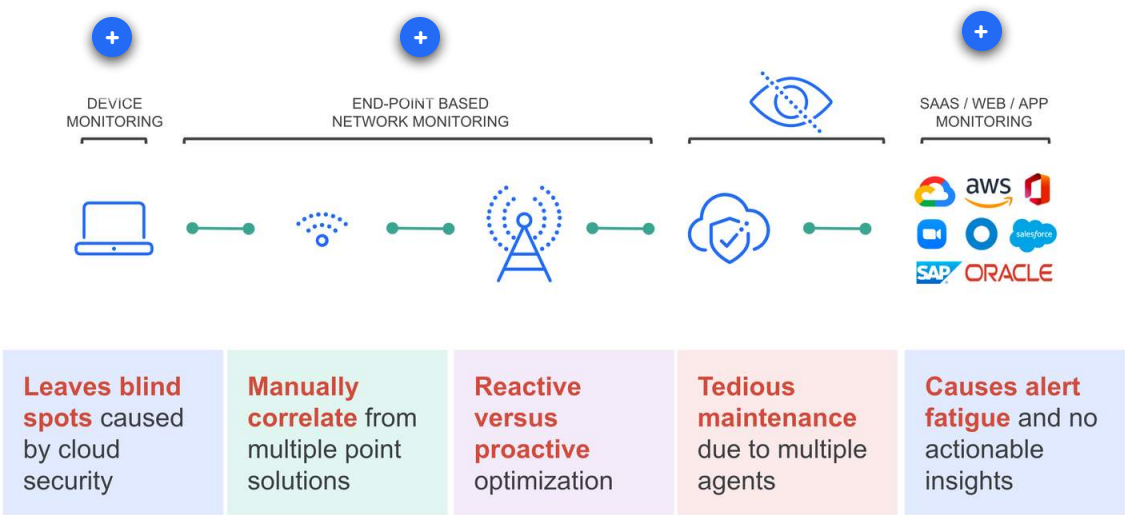
Current IT Monitoring Approaches Are Inadequate in a Work From Anywhere World

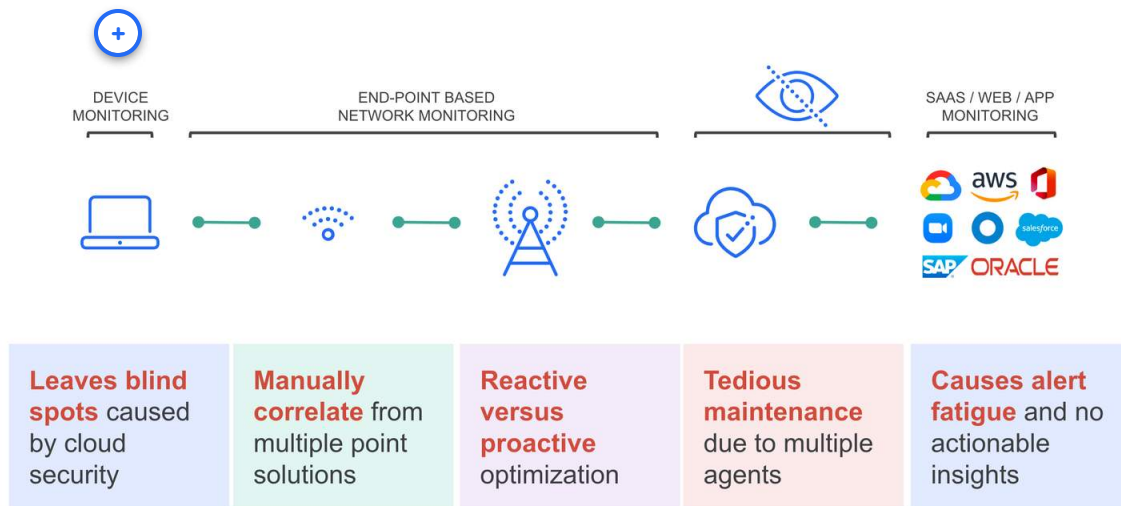
Most businesses today continue to have these multiple-point monitoring tools bought and managed by different IT teams. These tools create information silos and do not share any

These point monitoring tools optimized for data centers leave **visibility gaps** for detecting, troubleshooting, and diagnosing end-user performance issues across the internet. **Click**

context between them, leading to **fragmented visibility** into user experience and **extended troubleshooting time**.

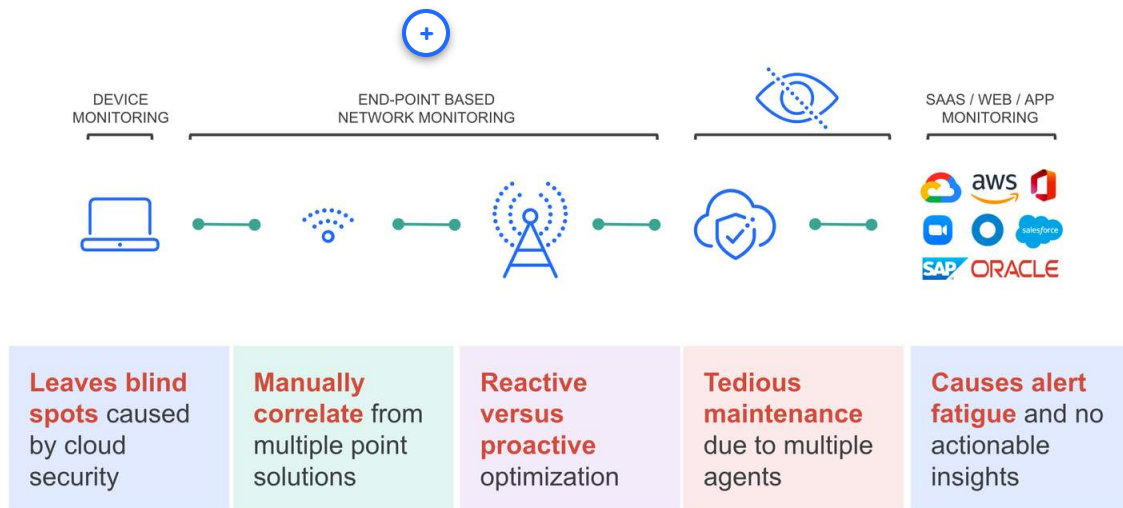
on each marker below to see what monitoring tools each team traditionally deployed and relied on.





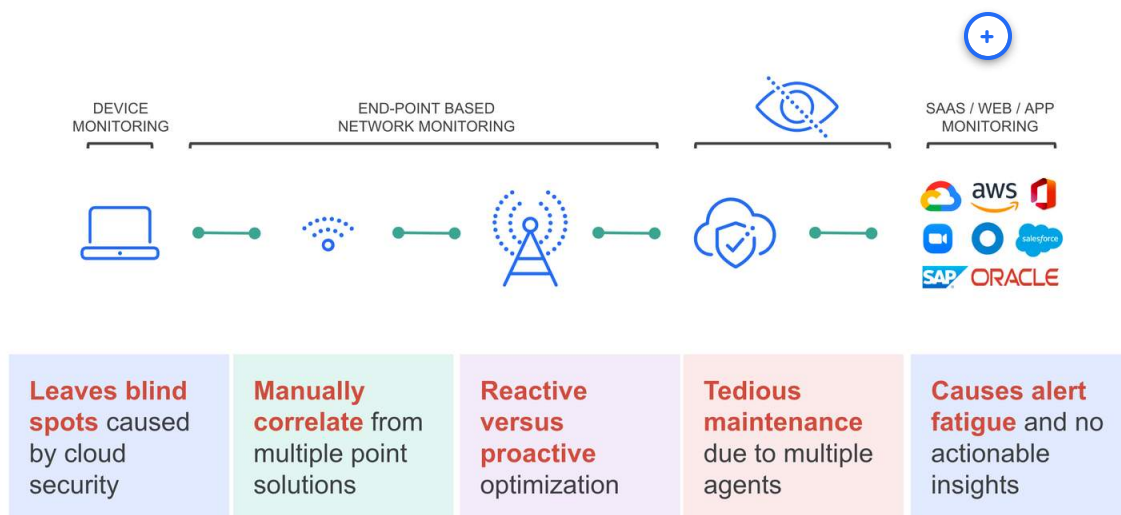
Device Monitoring

Service Desk/IT Help Desk Teams relied on broader IT Infrastructure and Device monitoring tools that relied on intelligence from various other tools to take action.



End-Point Based Network Monitoring

Networking teams deployed their own network monitoring tools, monitoring links, and health of network equipment with Simple Network Management Protocol (SNMP).



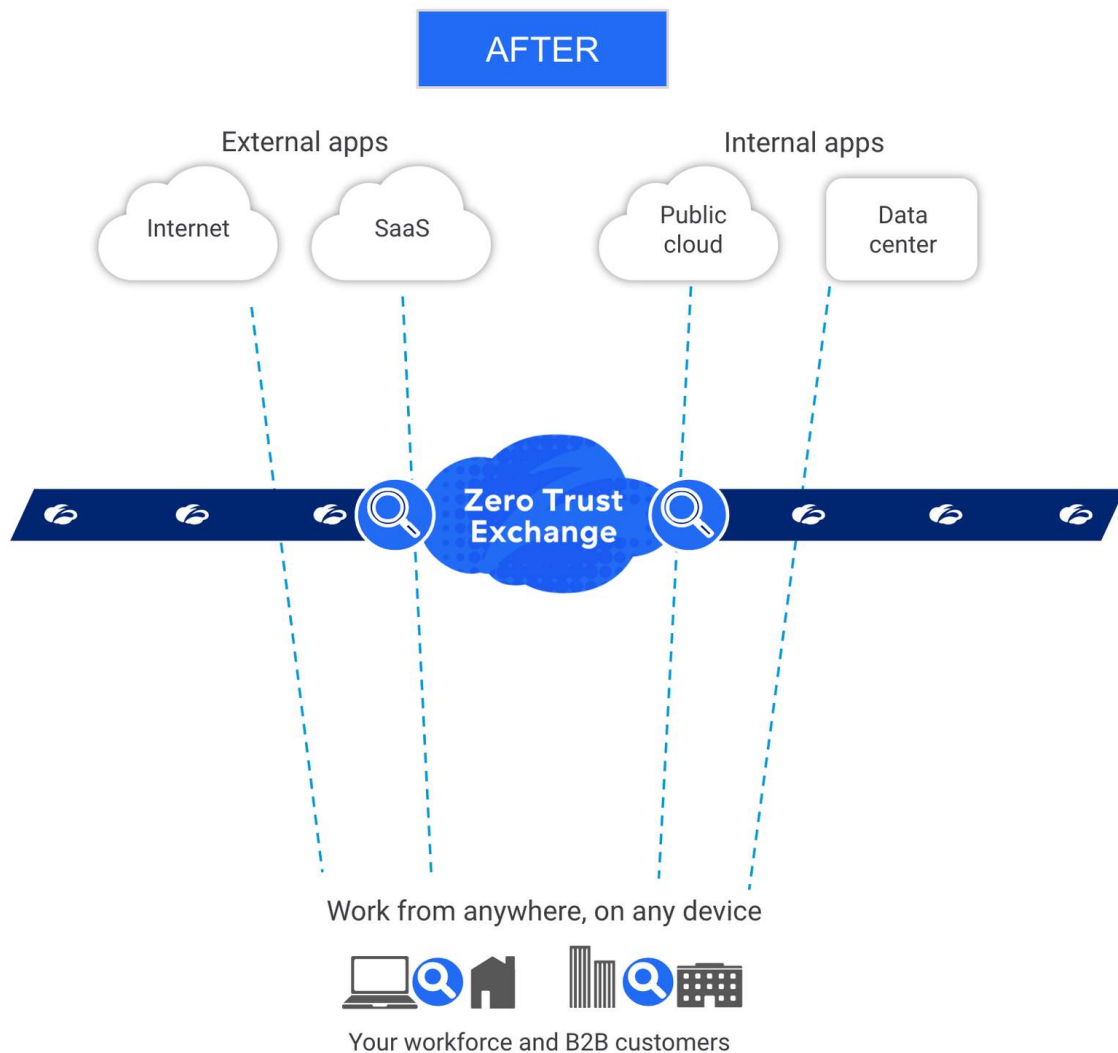
SAAS / WEB / APP Monitoring

Application teams would deploy Application Performance Management (APM) tools. As apps moved to the internet, they started relying on SaaS monitoring tools that came packaged with SaaS services. These included M365 and Teams monitoring tools, as well as trust portals.

CONTINUE

A New Monitoring Approach Is Required

In today's world, enterprise, network, and security are transforming to be aligned with a **zero trust driven architecture** where there is no backhaul or centralized security stack.



Organizations can no longer rely on traditional monitoring tools that take a data center-centric approach to monitoring and collecting metrics from fixed sites rather than directly from the user device.

Instead, **a unified monitoring system** is needed that views performance based on a user device, network path, or application whether the user is in the office, at home, or at a coffee shop.

Enter The World Of Digital Experience Monitoring



Digital Experience Monitoring (DEM)

Click here for a transcript of the video. —

With the recognition of new monitoring requirements in a cloud and mobile world, Gartner introduced digital experience monitoring, or DEM. A holistic approach to monitoring end-user experience without relying on legacy siloed monitoring tools such as endpoint-specific monitoring and network performance monitoring.

In the new world, the siloed tools should be consolidated into a holistic system that monitors three distinct variables that could impact an end user's experience: Network, Endpoint, and Application.

First, with the network, the goal is to look at every single hop that a user's traffic is taking from their device all the way to the destination and determine if there's an issue in the data path of which the user connects to that application. For example, if the service provider is sub-optimally routing internet-destined traffic, or experiencing a last mile provider issue.

Secondly, in today's work-from-anywhere world where everything is an application, it's important to monitor and visualize endpoint device events and system health metrics, including CPU, memory utilization, wifi signal strength, and more.

The third key aspect is the application. Unlike apps and the data center that can be instrumented and monitored with an APM tool, you cannot embed monitoring on SaaS applications. It's important to continuously monitor the health of SaaS applications and their uptime.

The end goal is to correlate network, endpoint, and application telemetry together in order to provide an effective view of an end user's experience, which can then be rolled up and aggregated at a location or at an organization level. From this new way of thinking stemmed the creation of Zscaler Digital Experience.

It's providing end-to-end visibility and troubleshooting of end-user performance issues for any user or application, regardless of location, while enabling continuous monitoring for Network, Security, Application, and Help Desk teams, with insight into end-user, device, network, and application performance issues.

CONTINUE

What Does ZDX Do?

How Zscaler Digital Experience Works

In the previous section, you heard about the background of Zscaler Digital Experience (ZDX) and the fundamental reasons for its existence.

Next, we'll explore what ZDX does that allows for continuous monitoring and measurement of digital experiences from users' devices.



ZDX secures fast and reliable access to applications from anywhere on any device through a scoring framework that provides always-on end-user experience monitoring.

In the video below, learn more about how ZDX works to provide endpoint, network, and application visibility to address two important goals.



How ZDX Works

Click here for a transcript of the video. —

Two major goals that ZDX works to address through the power of the Zero Trust Exchange and its integration with Zscaler Internet Access and Zscaler Private Access include:

1. Detecting issues that impact user experience
2. Reducing meantime to resolution
3. Keeping employees productive no matter where they are along with the ability to monitor UCAS applications.

With ZDX, organizations are able to understand how their traffic traverses the network, understand their hop-by-hop network performance, and since we take them to the front door of the application, they know the performance of the underlying application. This helps identify application versus network issues for critical applications quickly.

Based on the multiple telemetry points that we collect. We compute a score known as a ZDX Score. This can be good, bad, or okay for every user, application, location, or your entire organization as a whole.

The ZDX Score is computed based off of three variables: Endpoint, Network, and Application, and given a score between a value from 0 to 100. When the score is bad you can look at underlying

metrics to determine what is causing the bad score. The ZDX Score also provides a real-time indication of a systematic problem or a localized problem, which then can be diagnosed further using some of the troubleshooting capabilities in ZDX.

To monitor application performance, ZDX collects layer seven metrics such as server response time, DNS resolution time, and Page Fetch Time. From the network path, ZDX not only discovers each hop along the data path but also collects latency and packet loss values for each hop.

With ZDX, we can also collect metrics about device performance, such as CPU, memory, battery, disk utilization, and device events. For example, with Microsoft teams call quality monitoring, Zscaler will provide a Teams call quality score for your entire organization.

In this example, it's 85 out of 100, which is good. However, taking it to another level, we also provide the score for individual users. In this example, we are showing the call Quality is 17 for a specific user for this meeting that he or she has attended.

We can further drill down into the audio quality, video quality, and screen-sharing quality on that specific meeting. The ability to go back to a certain timestamp and see what was going wrong for that user is extremely powerful. And another example you can see in this diagram is a CEO's laptop. The CPU memory, disk utilization, everything looks good.

Then we take a look at the network path and we see that in this case between the hotel gateway and the ISP, there's high latency. The ability to pinpoint this quickly and take proper action is extremely important.

CONTINUE

Click each card below to review the core capabilities of Zscaler Digital Experience.

A dark blue square with a pattern of lighter blue circles of varying sizes. In the center, the text "Application Monitoring" is written in white, bold, sans-serif font.

Application Monitoring

ZDX provides active monitoring of application availability and uptime from the end-user device. Organizations can track critical performance metrics, including page fetch time (PFT) and server response time.

A dark blue square with a pattern of lighter blue circles of varying sizes. In the center, the text "Network Analytics" is written in white, bold, sans-serif font.

Network Analytics

ZDX and its CloudPath capability measures and analyzes end-to-end and hop-by-hop network path metrics from every user device to the cloud application, including Zscaler services such as ZIA and ZPA.



Endpoint Monitoring

ZDX tracks device health metrics, including Wi-Fi signal strength, CPU, memory usage, and network bandwidth usage for each user. Organizations can analyze device events and device metrics to gain insights into the overall health and performance of end-user devices.



UCaaS Monitoring

ZDX quickly identifies meetings and users that are experiencing poor performance in comparison to others using the ZDX score for call quality. Track mean opinion score (MOS) for each meeting participant all in a single dashboard.

Which core capability provides hop-by-hop network performance analysis?



Application Monitoring

- ☐ ZDX Scoring
- ☐ Network Analytics
- ☐ Endpoint Monitoring
- ☐ UCaaS Monitoring

SUBMIT

CONTINUE

Enabled Through Zscaler Client Connector

Another requirement and benefit of ZDX is its deployment through the Zscaler Client Connector. Instrumentation of ZDX starts at the Zscaler Client Connector, a unified agent for cloud security (ZIA), zero trust application access (ZPA), and digital experience monitoring (ZDX).

Single end-point agent for ZIA/ZPA/ZDX

Eliminates agent sprawl (FW, VPN), simplifies IT

Client Connector Forwarding Modes

Tunnel 1.0 and 2.0 (2.0 preferred)

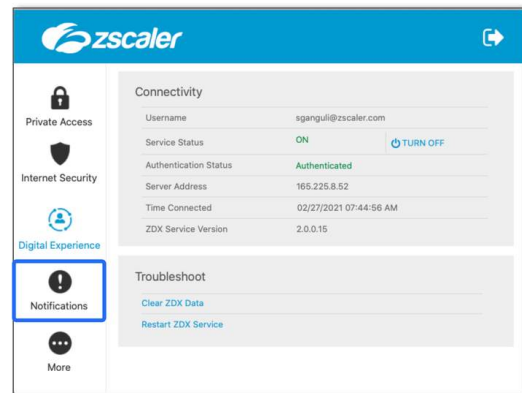
Lightweight and Extensible

Heavy lifting done in the Zscaler cloud, better device UX

Only Win and MAC for ZDX - no Android, iOS, or Linux support

Easy Deployment

Just enable license for ZDX in Mobile Portal - no new deployment



i For all Client Connector supported platforms and versions, visit our [Help Documentation](#).

Deploying ZDX has zero impact on end-users. The moment a license for ZDX is enabled, the client has a ZDX module that gets activated with <.5% impact on system resources.

It can be deployed for an entire user fleet or group-by-group from Active Directory by simply flipping a switch in the Zscaler Client Connector Portal.

CONTINUE

Critical Capabilities

You just observed how ZDX securely enables the ability to work from anywhere and ensures great digital experiences through its scoring system based on application monitoring, network analytics,

and endpoint monitoring. All of this is turned on with the flip of a switch on the Zscaler Client Connector.

Click each box below to view a full list of the additional critical capabilities ZDX provides.

AI-powered Root Cause Analysis —

ZDX automatically isolates root causes of performance issues, enabling less time for troubleshooting, eliminating finger-pointing, and getting users back to work faster.

Zscaler provides an ML-based analysis Y-Engine capability that analyzes the ZDX score over time and identifies which factors are impacting the score. The Y-Engine then provides suggested areas to investigate.

Software & Device Inventory & Metrics —

Companies can fully understand their software portfolio and versions deployed across their organization and on each device. They can rapidly troubleshoot and fix end-user device issues without having to remote in, and keep them in compliance.

ISP Insights —

ZDX monitors the Health of the Internet allowing organizations to be the first to spot ISP incidents across the globe, by severity. Customers can pick top-performing ISPs to optimize user experience. Learn more about ISP Insights [here](#).

Pre-defined Templates —

Organizations can utilize pre-defined application monitoring templates such as Sharepoint Online, Outlook Online, MS Teams, Zoom, Salesforce, and ServiceNow to rapidly deploy and collect telemetry data.

Live Troubleshooting —

ZDX provides a Deep Tracing capability that allows organizations to triage user performance issues in near real-time and pinpoint the root cause at the end-user device, network path, or cloud application.

Robust API Integrations —

Organizations can integrate ZDX digital experience insights with popular ITSM tools like ServiceNow to provide additional insights and trigger remediation workflows.

Alerting —

Organizations can set up real-time alerts and customize them to meet their performance needs. ZDX integrates easily with a company's service management tools such as ServiceNow and pushes notifications through webhook or email.

CONTINUE

Initiatives: Why ZDX Is Important

Top Initiatives Driving Adoption

Now that you understand how ZDX works and its critical capabilities, let's dive into the value the product provides. Did you know?

>63%

of employees polled prefer hybrid or remote work

McKinsey

93%

of CIOs are moving or have moved to SaaS

Deloitte

>35%

more helpdesk tickets are opened in hybrid workplaces

DeepCoding

In response to these statistics, Zscaler has worked to ensure ZDX provides maximum value to organizations as they navigate this

When identifying ZDX opportunities, it is important to listen and watch out for the following top initiatives that organizations may be looking to

new hybrid workforce of experience
monitoring.

address on their secure digital transformation
journey.

- Work From Anywhere
- Modern Workplace Enablement
- M365 Performance Optimization
- End-User Experience Monitoring
- Reduce Mean Time to Detect (MTTD) and Mean Time to Resolve (MTTR)
- Reduce Total Number of Trouble Tickets

CONTINUE

Pains ZDX Solves For

The Pains ZDX Solves For

Now that we've addressed top initiatives, let's review the challenges and consequences many organizations are facing as a result of not having ZDX.

It's important to be reminded of these challenges in order to discuss the value and benefits ZDX provides.

Click each card below to view each challenge and its consequence.



Loss of Productivity

Challenge

Poor user experience due to slow and unreliable connectivity to applications.

Consequence

Lack of proactive insights into poor application and end-user device performance issues leads to lost productivity and revenue



Impaired Agility

Challenge

Lack of unified view into user experiences – user device, network path, and application together slows down IT teams.

Consequence

Elongated time to identify and resolve user experience issues leading to ticket pile-ups.



Complexity & Costs

Challenge

Costly point monitoring solutions do not provide end-to-end performance insights from the user to the application.

Consequence

Troubleshooting becomes a manual process of correlating information from various tools.

CONTINUE

Why are organizations paying high costs for their current monitoring tools?

☐

They have no ability to identify, control, and govern sensitive data at rest

☐

They have siloed point monitoring solutions

☐

They have no visibility into the data path

SUBMIT

CONTINUE

Benefits & Differentiation of ZDX

What ZDX Brings to the Table

How does ZDX address these challenges? Why is ZDX needed right now?

There are three key benefits that Zscaler Digital Experience provides. **View each of the key benefits below** and then **watch the short video** to hear how ZDX achieves each benefit.



Ease of Use, Tools, & Operational
Costs



Business Productivity, Agility, &
Innovation



Visibility, Analytics, and Reporting



ZDX Benefits

Click here for a transcript of the video. —

Let's expand on the benefits of ZDX in order to see how valuable the offering is for customers, starting with the ease of use, tools, and operational costs.

First, with ZDX you have a simplified administration with full Role-Based Access Control built into the product. This enables multiple teams across your org to access ZDX data, but only the information required for their role. Additionally, with a single pane of glass and the consolidated approach decreases the time to triage events across endpoint, network, and application all while providing more meaningful alerts with less false positives.

For example, let's say you're experiencing issues with Microsoft Teams. How does your support team diagnose if the cause of poor voice quality was from the endpoint's high CPU utilization, the team's version, low wifi, signal, or an issue with a local ISP? Tying these data points together lowers the amount of tools required in escalations for an organization.

Let's next expand on the benefit of business productivity agility and innovation. As you think about measuring end-user experience, we know ZDX gives you a measurable metric driven by ZDX Score. But we also collect a lot of underlying raw data points to understand exactly where the problem is.

In addition, when you think about moving to the cloud and adopting new cloud services, one of the key factors is to understand how users will perceive the value of the transformation and how you will solve problems when they run into issues.

This leads us into the third benefit of visibility, analytics, and reporting. Having a better recovery time for when things go wrong is very important. Every five minutes we're able to collect and see the stats across all of your users from different parts of the globe. With ZDX there's also granularity right up to the user's level. You can determine the root cause of issues with Automated Analysis for users that have been impacted in the last fourteen days without requiring another tool or waiting for the problem to reoccur.

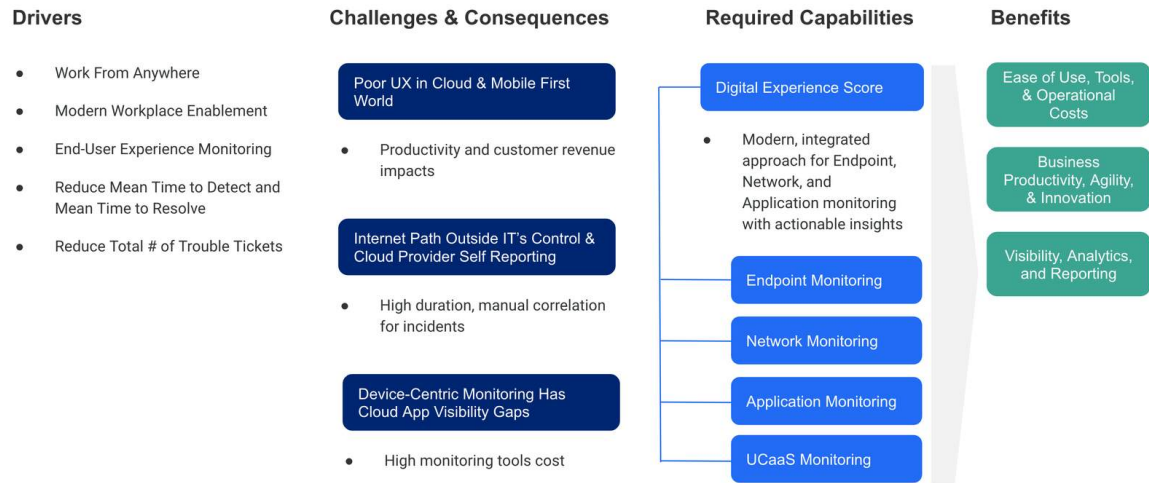
The Unique Differentiation of ZDX

With an integrated digital experience monitoring service, ZDX and Zscaler's Zero Trust Exchange provides unique differentiation from the rest of the industry in the following ways:

- **Increase agility** and collaboration among Application, Security, Network, and Helpdesk Operations teams, while triaging and resolving user experience issues.
- **Improve productivity** with better user experience and fast, secure, and reliable connectivity through the Zscaler Cloud.
- **Reduce complexity and cost** of point monitoring solutions.
- **Simplify operations** using the same lightweight agent for all Zscaler services.

CONTINUE

From Drivers to Benefits



Use this image as a reference in understanding the total value of ZDX, mapping the drivers all the way to the benefits discussed in this module.

CONTINUE

ZDX Dashboard Overview



ZDX Live in Action

Up to this point, we have explored what Zscaler Digital is, the powerful capabilities it provides, and the critical business initiatives it addresses.

Let's now take all of this information and apply it to how ZDX operates within the Administrator portal.

In this section, you will be taken on a guided tour of the main dashboards, allowing you to familiarize yourself with the administrative portal and how it can address customers' needs.



This demonstration is showcased using Zscaler's SafeMarch environment. Capabilities are continuing to be built out in the Solutions Demo Center for you to

utilize with customers.

ZDX Dashboard Overview



Click here for a transcript of the video. —

Let's begin by taking a quick look at the Zscaler Digital Experience or ZDX Administration portal, where administrators manage and interact with the service.

As you can see the layout here in the Admin portal is pretty straightforward. On the left-hand side, we have a vertical navigation bar where an administrator finds and interacts with all the different areas of functionality within ZDX.

At the top of the navigation bar, we find the ZDX, Dashboard, which provides a high-level overview of all user activity globally along with summary statistics reflecting heavily used applications and services. Note that the dashboard includes a global summary of the ZDX Score. In short, the ZDX Score is a carefully calculated metric ranging from 1 to 100, which reflects the overall quality of any particular user's experience. ZDX Scores are a fantastic way to quickly understand whether a user has experience in a given application or service is good, okay, or poor. And of course, an admin can quickly drill into a ZDX score, to identify all the factors that contribute to it.

The Applications panel provides a comprehensive breakdown of the cloud apps and services which users are accessing, whether those applications are predefined or custom.

The User's panel gives an overview of user activity and experience scores associated with whichever applications are selected.

The User Search tool allows an administrator to quickly jump to a particular user just by entering their name or email address.

The Inventory option is where admins can quickly view comprehensive inventories of their managed endpoints. First here in the software overview panel, an administrator can quickly view a beautiful, easy-to-understand visual summary of all the different applications in use on their user's endpoints with lots of filtering options available if needed. Similarly here in the device overview panel, an administrator can see a similar breakdown of all the different endpoints being used by their users, whether those devices are physical or virtual.

The Configuration panel is where an administrator can quickly configure ZDX monitoring for applications by selecting from predefined apps or by creating their own custom applications, once an application has been selected or created. The administrator will configure probes, which are automatic performance and connectivity tests that help monitor each application.

The Administration menu contains some behind-the-scenes configuration tools such as adding additional admin accounts, creating roles with different levels of access and functionality within the portal, and more.

Finally, the Alerts menu allows admins to set customized alerts. So the right people can be automatically notified when specific criteria are met. For example, if a critical cloud application starts performing poorly, or if a user's endpoint becomes overloaded, an alert can be automatically sent to the appropriate team so they can take corrective action. Given that more and more users are working remotely every day, this level of insight into the complete user experience is tremendously valuable to organizations seeking to maximize their user's productivity.

As you've seen the ZDX portal is a single pane of glass where an organization can configure, monitor, and analyze many factors that can contribute to the overall productivity and online experience of their users. Thanks for following along with this guided tour.

CONTINUE

Recap



In this module, you learned about Zscaler Digital Experience (ZDX), one of Zscaler's core offerings that work through the Zero Trust Exchange to ensure great user experiences across all locations, devices, and applications. ZDX allows organizations to:

- Get visibility across devices, networks, and apps—even those not in their control
- Detect issues before users complain, and quickly pinpoint the root cause
- Simplify their monitoring stack with a single end-to-end view

Zscaler ZDX Website

Explore the ZDX page on our corporate website to learn more about what the offering can do for organizations, view its plans and features, and dive into additional resources, demos, and customer case studies.

[GO TO WEBSITE](#)

From this module, you should now be able to:

1

Provide an overview of the background of Zscaler Digital Experience, along with its core components and critical capabilities

2

Describe the value of Zscaler Digital Experience, including the initiatives it drives, pains it addresses, and benefits it provides

3

Identify the main key differentiators that uniquely set ZDX apart from other industry solutions

CONTINUE

Thank You!